Candidate Seat no………….

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Second Semester of M.Tech (EC) (Embedded System &VLSI Design) Examination**

**May 2018**

**EC752/EC739 VLSI Design Verification & Testing**

**Date: 08.05.2018, Tuesday** **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

| | | | |
|---|---|---|---|
| **Q - 1** | | **Answer the question below.** | **[07]** |
| | **a.** | What is System Verilog? What are the advantages of System Verilog? | **[03]** |
| | **b.** | Differentiate System Verilog and System C. | **[02]** |
| | **c.** | What is the importance of Verification? How to generate clock in Verilog? | **[02]** |
| **Q – 2** | | **Attempt any two questions.** | **[14]** |
| | **a.** | Draw Block Diagram of Test bench Architecture. Explain each block in brief. | **[07]** |
| | **b.** | Write a Verilog Code and Test bench for Vending Machine Example with necessary assumptions. (1= Cold Drink, 2= Tea,3=Coffee,4= Exit). | **[07]** |
| | **c.** | Explain Different Types of Structural Faults. | **[07]** |
| **Q – 3** | | **Attempt any two questions.** | **[14]** |
| | **a.** | Design Gate Level Verilog Code for R-S latch and J-K Flip-flop. | **[07]** |
| | **b.** | Write Verilog Code for 8-bit Arithmetic Logic Unit (ALU). | **[07]** |
| | **c.** | Explain Gate level and System level functional faults with necessary examples. | **[07]** |

## SECTION – II

| | | | |
|---|---|---|---|
| **Q - 4** | | **Answer the question below.** | **[07]** |
| | **a.** | What is $time in Verilog? What is the difference between $monitor and $display in verilog? | **[03]** |
| | **b.** | What is the sensitivity list in verilog? Write a Verilog Code for Synchronous and Asynchronous Reset. | **[04]** |
| **Q – 5** | | **Attempt any two questions.** | **[14]** |
| | **a.** | Explain Automatic random test pattern generation D-algorithm for combinational circuit with suitable example. | **[07]** |

**b.** Explain different types of Data Types and different types of Operators of Verilog. **[07]**

**c.** Explain Universal Verification Methodology (UVM) for verification of Device Under Test (DUT) with block diagram. **[07]**

**Q – 6** **Attempt any two questions.** **[14]**

**a.** Define following terms: Test Cycle, Test Round and Test Session. Explain One's Counter with necessary Verilog Code. **[07]**

**b.** Explain Ring Counter and Twisted Ring Counter for Memory Testing. **[07]**

**c.** Draw Block Diagram of Separate BEST (Built In Evaluation and Self Test) BIST (Built in Self Test) Architecture and RTS (Random Test Socket) BIST (Built in Self Test) Architecture. **[07]**

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